

GENETIC ENGINEERING: GM CROPS AND CRISPR

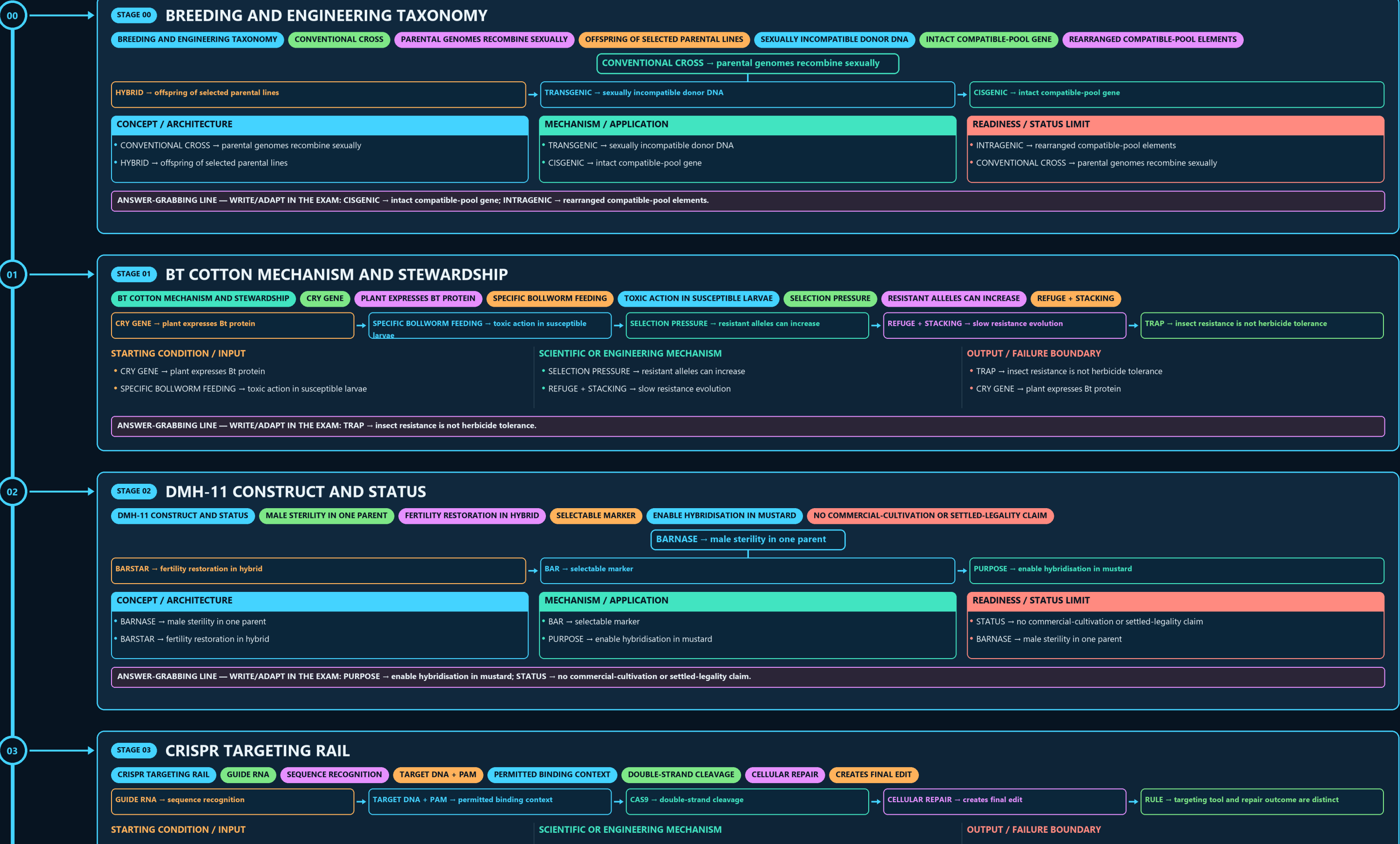
BREEDING AND ENGINEERING TAXONOMY → BT COTTON MECHANISM AND → DMH-11 CONSTRUCT AND STATUS → CRISPR TARGETING RAIL → EDITING-OUTCOME MATRIX → SDN CLASSIFICATION LADDER

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g1 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles



- BARNASE → male sterility in one parent

- SBCC + DLC → state and district monitoring
- RDAC → recombinant-DNA advice

06

STAGE 06 SIX-TIER BIOSAFETY INSTITUTION MAP

SIX-TIER BIOSAFETY INSTITUTION MAP

RECOMBINANT-DNA ADVICE

INSTITUTIONAL DAY-TO-DAY OVERSIGHT

RESEARCH AND CONTAINED STAGES

ENVIRONMENTAL RELEASE AND LARGE-SCALE USE

SBCC + DLC

STATE AND DISTRICT MONITORING

RDAC → recombinant-DNA advice

IBSC → institutional day-to-day oversight

RCGM / DBT → research and contained stages

GEAC / MoEF&CC → environmental release and large-scale use

CONCEPT / ARCHITECTURE

- RDAC → recombinant-DNA advice
- IBSC → institutional day-to-day oversight

MECHANISM / APPLICATION

- RCGM / DBT → research and contained stages
- GEAC / MoEF&CC → environmental release and large-scale use

READINESS / STATUS LIMIT

- SBCC + DLC → state and district monitoring
- RDAC → recombinant-DNA advice

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SBCC + DLC → state and district monitoring.

07

STAGE 07 LAW AND REGULATOR SPLIT

LAW AND REGULATOR SPLIT

RULES 1989

CORE GMO BIOSAFETY FRAMEWORK

CONTAINED WORK

ENVIRONMENTAL APPRAISAL

GM-FOOD SAFETY APPROVAL AND LABELLING TRACK

CROP SYSTEM

AGRONOMIC EVALUATION AND VARIETY PROCESSES

SYSTEM / DEVICE / INSTITUTION

RULES 1989 → core GMO biosafety framework

RCGM → research / contained work

DECISIVE TECHNICAL DISTINCTION

GEAC → environmental appraisal

FSSAI → GM-food safety approval and labelling track

STATUS / UNIT / EVIDENCE LIMIT

ICAR / CROP SYSTEM → agronomic evaluation and variety processes

RULES 1989 → core GMO biosafety framework

SYSTEM / DEVICE / INSTITUTION

- RULES 1989 → core GMO biosafety framework
- RCGM → research / contained work

DECISIVE TECHNICAL DISTINCTION

- GEAC → environmental appraisal
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STATUS / UNIT / EVIDENCE LIMIT

- ICAR / CROP SYSTEM → agronomic evaluation and variety processes
- RULES 1989 → core GMO biosafety framework

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ICAR / CROP SYSTEM → agronomic evaluation and variety processes.

08

STAGE 08 INDIAN DECISION TIMELINE

INDIAN DECISION TIMELINE

09 FEB 2010

BT BRINJAL MORATORIUM

30 MAR 2022

SDN-1/2 EXEMPTION OM

17 MAY 2022

DBT GENOME-EDITED PLANT GUIDELINES

23 JUL 2024

09 FEB 2010 → Bt brinjal moratorium

30 MAR 2022 → SDN-1/2 exemption OM

17 MAY 2022 → DBT genome-edited plant guidelines

23 JUL 2024 → Supreme Court DMH-11 split verdict

04 APR 2025 → two genome-edited rice varieties announced

CONCEPT / ARCHITECTURE

- 09 FEB 2010 → Bt brinjal moratorium
- 30 MAR 2022 → SDN-1/2 exemption OM

MECHANISM / APPLICATION

- 17 MAY 2022 → DBT genome-edited plant guidelines
- 23 JUL 2024 → Supreme Court DMH-11 split verdict

READINESS / STATUS LIMIT

- 04 APR 2025 → two genome-edited rice varieties announced
- 09 FEB 2010 → Bt brinjal moratorium

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: 23 JUL 2024 → Supreme Court DMH-11 split verdict.

09

STAGE 09 BIOSAFETY MECHANISM MAP

BIOSAFETY MECHANISM MAP

TARGET PEST PRESSURE

RESISTANCE EVOLUTION

SEED MOVEMENT

GENE-FLOW EXPOSURE

TRAIT EXPRESSION

NON-TARGET EXPOSURE PATHWAY

SIMILAR DNA SITES

TARGET PEST PRESSURE → resistance evolution

POLLEN / SEED MOVEMENT → gene-flow exposure

TRAIT EXPRESSION → non-target exposure pathway

SIMILAR DNA SITES → possible off-target edit

STARTING CONDITION / INPUT

- TARGET PEST PRESSURE → resistance evolution
- POLLEN / SEED MOVEMENT → gene-flow exposure

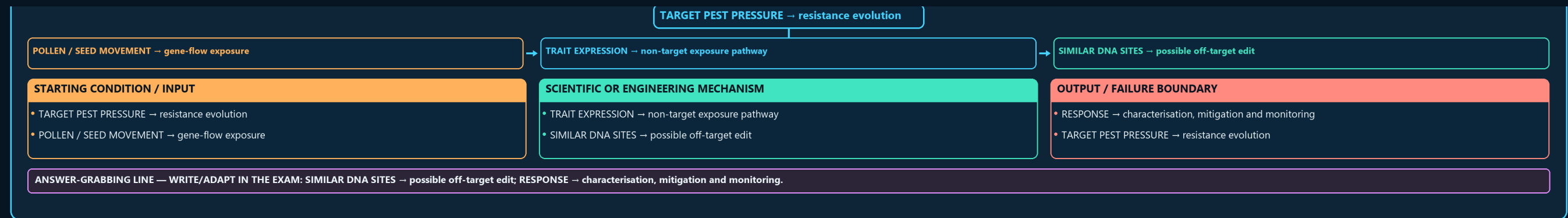
SCIENTIFIC OR ENGINEERING MECHANISM

- TRAIT EXPRESSION → non-target exposure pathway
- SIMILAR DNA SITES → possible off-target edit

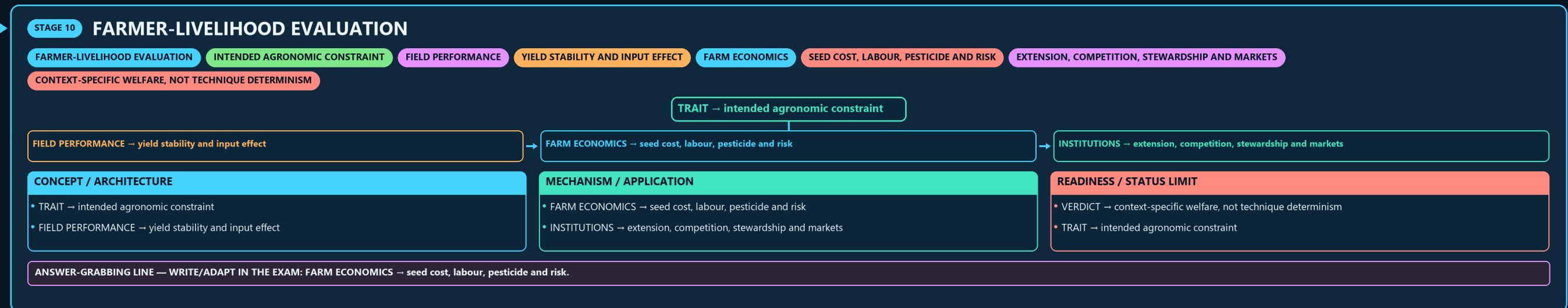
OUTPUT / FAILURE BOUNDARY

- RESPONSE → characterisation, mitigation and monitoring
- TARGET PEST PRESSURE → resistance evolution

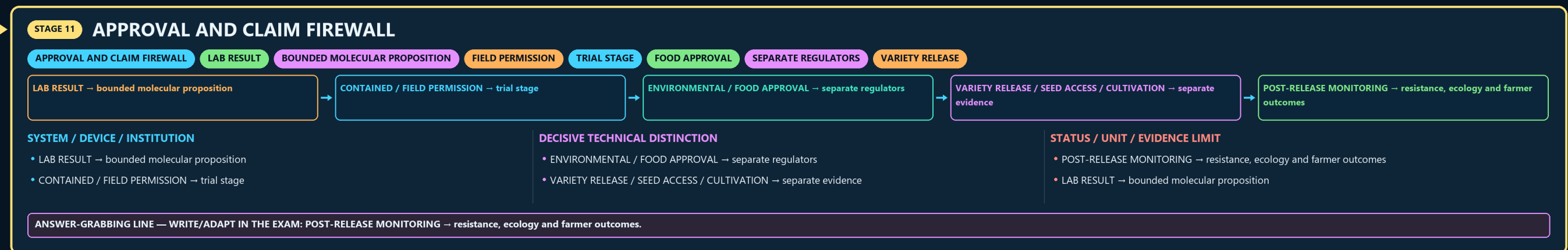
ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SIMILAR DNA SITES → possible off-target edit; RESPONSE → characterisation, mitigation and monitoring.



10



11



E

